

Ideology or What? Legislative Behavior in Multiparty Presidential Settings

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In this paper I show that voting patterns of Brazilian legislators depart from their ideology in ways that suggest that the president plays an important role in influencing their behavior. Moreover, statistical analysis indicates that this influence is channeled through the distribution of pork and nominations to cabinet positions. Ideology not only fails to fully explain the patterns of legislative behavior observed since the return to democracy, but there is evidence that ideological behavior has declined over time.

Latin American parties can be arrayed along a left-right scale based on reputation or platform like parties elsewhere, but legislative voting is determined as much by whether legislators receive tangible benefits from the president as it is by ideology. This fact, which finds both anecdotal and systematic support, has important implications for political analysis in general, and executive-legislative relations in particular. If ideology is not the main thrust behind how legislators behave, one must incorporate legislators' other concerns into the analysis.

While roll calls are an objective measure of legislative behavior, the president frequently exerts significant influence over how legislators behave. This should be especially true when an elected president coexists alongside a multiparty legislature and when the president has a monopoly or near monopoly over the distribution of political resources that are important to the legislators' future political careers. The president can sway legislators away from their ideology when the bureaucracy responds almost exclusively to the president, when the president can create, rearrange, and extinguish agencies and departments, make decisions regarding political appointments, and exert final say over the time, manner, and even place of government expenditures. These elements are frequently found in many Latin American countries but are conspicuously present in Brazil.

In this paper, I use data from Brazil to explore the differences between legislators' ideology and their behavior in multiparty presidential systems. I present

a measure of legislators' ideology and then analyze the patterns in which legislative behavior—as revealed by the analysis of roll-call votes in the lower house—deviates from it. The comparison between ideology and behavior is compatible with the claim that the president's deliberate actions, such as the distribution of pork and cabinet positions, contribute alongside ideology to determining how legislators behave. Not only does ideology not fully explain the observed patterns of legislative behavior, but the effect of ideology has declined over time—contrary to what might have been expected as the party system in a new democracy became consolidated. This paper dwells on evidence from a single country, and only future research will show the extent to which these substantive findings are generalizable. Nevertheless, the paper's most general point—that prior knowledge about the political processes that generate roll-call data is crucial in determining how it should be analyzed and interpreted—is broadly applicable.

I start with a simple description of legislative behavior in Brazil that suggests a mismatch between behavior and ideology. Subsequently, I present a method for obtaining estimates of ideology exogenous to legislative behavior and explore the patterns by which observed behavior deviates from these estimates. I then perform a statistical analysis of the determinants of legislative behavior and interpret the results. The conclusion summarizes the argument and points toward ways in which the findings could be generalized to other cases.

Legislative Behavior in Brazil

Most depictions of Brazilian party politics stress the extremely low levels of party identification in the electorate, the fuzziness of the policies defended by most parties, the wide variation in political culture and coalition patterns across states, and the frequent and pervasive party switching by politicians at all levels (Ames 2001; Desposato 2006; Mainwaring and Scully 1995; Mainwaring and Shugart 1997; Shugart and Carey 1992; among others). While there is considerable truth in these observations, Brazilian politics displays a much more nuanced combination of volatility and stability.

The seminal work of Figueiredo and Limongi (1999) shed light on the inner workings of the Brazilian legislature, showing that parties behave much more coherently than the casual observer would think, at least within parliament. While the electoral system is extremely permissive, institutions within Congress help routinize and structure political practices in the legislature. More recently, Lyne (2008) has argued that a more active leadership has led parties to be more cohesive in the recent democratic period compared to the country's previous stint with democracy from 1946 to 1964. But if, in fact, parties do behave in a structured way within the legislature, does it necessarily follow that this structure is *ideological*? In other words, what role does ideology play in executive-legislative relations, and consequently, in the functioning of the legislative branch?

The analysis of the voting behavior of legislators provides a first glimpse into the matter. Table 1 shows one-dimensional W-Nominate ideal-point estimates,¹ by party, for the last five legislatures in Brazil.² For the 48th and 49th Legislatures, the snapshot of legislative behavior corresponds quite well to what would be the perceived ideological ordering of

Brazilian parties.³ From the 50th Legislature onwards, however, the PFL (right) and the PSDB (center)—the core of former President Cardoso's coalition and today the core of the opposition to President Lula's government—appear clustered at the right end of the scale, greatly overstating the reputed ideological distance between the PSDB and Lula's party, the leftist PT. One could be tempted to interpret this new ordering simply as the product of a "move to the right" by the PSDB, since in general terms other parties considered leftist still appear on the left of the scale, and vice-versa. However, the image of the 52nd legislature, which coincides with Lula's first term in office, is mind-boggling: the PSDB and the PFL are isolated in the far right of the scale, the leftist PDT and PPS appear on the right, and the rightist PP and PTB are on the left, close to the communist PC do B.

In trying to make sense of these results, it is important to note that ideal point estimates are based solely on the similarity of the legislators' voting records. Though single-dimension estimates are commonly interpreted as the traditional left-right ideological scale, their actual substantive meaning is entirely subjective. Additionally, the determination of what is "left" and what is "right" is also an arbitrary decision, and as I will show, not an entirely obvious one.

If one entertains the possibility that the retrieved dimension is not ideology, what could it be? A slightly more careful examination of Table 1 suggests that instead of an ideological order, parties are ordered roughly in a government-opposition dimension.⁴ Evidence of this can be seen through an analysis of the positions of specific parties. Take the case of the leftist PDT, for instance, and its position relative to the also leftist PT. Despite style differences, both parties historically displayed similar parliamentary behavior, and ideological orientations, and entered into many electoral alliances in the past. In the 2002 elections, the PDT endorsed Ciro Gomes (then in the PPS) in the first round of the presidential election but supported Lula in the second round and

¹A considerable literature exists on W-Nominate itself (Lewis & Poole 2004, Poole 2005, Poole & Rosenthal 1985) and on other approaches to estimate ideal points (Clinton, Jackman & Rivers 2004, Londregan 2000, Krehbiel & Rivers 1988). Ideal point estimates used in this paper were obtained using W-Nominate for R (Poole, et al. 2007).

²Data are only analyzed for the lower house (*Câmara de Deputados*), where legislators are elected for four-year terms, by an open-list proportional system, with states serving as electoral districts, and magnitude currently ranging from 8 to 70. The 48th legislature served from 1987 through 1990, but only data from the post constitutional assembly period (1989–90) is used. Typically, close to 20 parties obtain representation in the lower house.

³I discuss the ideological organization of Brazilian parties in more detail later in the text, but for now it suffices to say that this ordering would be accepted by most observers of Brazilian politics (Alcántara 1994–2005, Coppedge 1997, Figueiredo & Limongi 1999, Kinzo 1993, Mainwaring & Pérez-Liñán 1997, Rosas 2005).

⁴Leoni (2002), the first to employ W-Nominate in the analysis of data from Brazil, had already hinted at this.

TABLE 1 Median Ideal Points by Legislative Session: Selected Parties

(a) 48 th Legislature (1987–1990)										
PC do B	PT	PPS	PDT	PSDB	PMDB	PTB	PFL	PL	PPB	PRN
−0.84	−0.81	−0.54	−0.46	−0.10	0.19	0.37	0.47	0.48	0.51	0.52
(b) 49 th Legislature (1991–1994)										
PT	PC do B	PDT	PPS	PSDB	PMDB	PTB	PL	PFL	PPB	PRN
−0.63	−0.46	−0.18	−0.17	0.12	0.34	0.56	0.63	0.66	0.69	0.69
(c) 50 th Legislature (1995–1998)										
PT	PC do B	PDT	PPS	PMDB	PL	PPB	PTB	PSDB	PFL	
−0.86	−0.75	−0.41	−0.33	0.29	0.30	0.36	0.44	0.46	0.60	
(d) 51 st Legislature (1999–2002)										
PC do B	PDT	PT	PPS	PL	PTB	PMDB	PSDB	PFL	PPB	
−0.80	−0.71	−0.64	−0.35	−0.12	0.12	0.15	0.37	0.38	0.44	
(e) 52 nd Legislature (2003–2007)										
PT	PL	PC do B	PTB	PPB	PMDB	PPS	PDT	PSOL	PSDB	PFL
−0.62	−0.32	−0.28	−0.27	−0.07	−0.05	0.03	0.15	0.29	0.59	0.67

Notes: Ideal points in one dimension were estimated using W-Nominate for R on roll call data provided by Limongi & Figueredo, and the *Camara de Deputados*. Median estimate for each party is shown under the party label, and parties are ordered from left to right by this value. Estimates for 48th Legislature rely on 1989 and 1990 data only. Point estimates are *not* directly comparable across years. Parties that changed names are represented by their acronym as of the start of the 52nd legislature.

was an early ally of the newly elected government. The PDT was also the first party to break with the government at the end of its first year in power. Though Table 1 suggests that the PDT moved to the *right* of Lula's PT, most would agree that after leaving the government coalition, the PDT occupied the space of opposition to the government from the *left* throughout the remainder of Lula's first term in office.

Even clearer is the situation of the PSOL. Commanded by Senator Heloísa Helena, this radical left party was formed by legislators who split or were expelled from the PT for voting against the government's pension reform. However, the party's location shown in Table 1(e) is well to the *right* of the PT and much closer to Cardoso's *center-right* PSDB, even though the *only* thing the PSOL and the PSDB had in common was their opposition to Lula's government.

My claim is that other factors beside ideology drive party behavior in the legislature. Consequently, the underlying dimension of conflict retrieved by W-Nominate estimates is not an ideological one. More concretely, I argue that the government seeks to gain legislative support by using different resources at its disposal, which gives rise to a government-opposition dimension of conflict. The dispensation of government resources can range from outright pork distribution, to more subtle things such as allowing legislators to claim credit for government initiatives. Still, it always involves the concession or transfer of

political resources—not necessarily material ones—that are originally under the control of the president, but which are important to further the careers of all politicians.⁵

In the remainder of the paper, I provide more evidence of this process. I also provide an outline of the theoretical underpinnings of my approach, though I leave a comprehensive exposition of the actual theory to another venue (Zucco Jr. 2007). Before proceeding, it is important to note that my argument is compatible with recent studies that have noted the important role the president plays in shaping legislative behavior. Pereira (2002) and his coauthors have focused on the importance of government handouts to the political strategies of individual legislators reinforcing the depiction originally put forth by Ames (2001) of the Brazilian Congress as populated by locally minded legislators. More recently, Santos (2006) has argued that legislators use their position relative to the president as a way to communicate with voters, and for this reason the president's influence contributes to nationalize—as opposed to “parochialize”—legislators's behavior. Though the two mechanisms are different, both stories share the view that the executive plays an important role in shaping legislative behavior.

⁵Though in this paper I focus on executive-legislative exchanges, I acknowledge that party decision making is more complex than this and is potentially affected by other factors that transcend the scope of this paper, such as intraparty dynamics, public opinion, and strategic electoral considerations.

Illustrating the Government-Opposition Dimension

The past few Brazilian governments have shared stable positions on core issues, such as taxes, interest rates, inflation, and minimum wage, irrespective of the nominal ideological inclinations of the incumbent president. For instance, Lula pushed for measures his party—the leftist PT—had fought against during its entire previous history, such as the taxation of retirement benefits. Conversely, the PSDB—Cardoso’s center-right party—has often switched sides relative to positions it defended when in government (e.g., the tax of financial transactions, known as the CPMF). Even the arguments used by either side have shifted and seem more an attribute of whether the party is in power or opposing the government. Governments point to budget constraints, while the opposition highlights the “needs” of beneficiaries and the “fairness” of their proposed policies.⁶

As an example of these “role contingent preferences,” I present a brief analysis of the legislative debates on the minimum wage bills in 2000—the second year of Cardoso’s second term—and 2004—the second year of Lula’s first term in office. Some characteristics of the minimum wage bill make it an interesting case study. First, it is debated regularly, thus allowing for comparisons over time. Second, much of the debate is about the nominal monthly value of the wage, which facilitates the comparisons of the revealed preferences of politicians. Finally, there are clear expectations regarding purely ideological preferences: the left should prefer a higher minimum wage than the right. The interesting twist is that a large portion of the benefits paid by Social Security is indexed by the minimum wage, so any increase in its value greatly affects government accounts. For this reason, the competing role-contingent expectation is that the government will prefer a lower minimum wage than the opposition, regardless of ideology. The minimum wage example, thus, allows for the comparison between role contingent and ideological preferences.

In February 2000, Cardoso preemptively sent Congress a bill setting the minimum wage at R\$ 151, but part of his support coalition threatened to unite with the opposition in support of an increase to

R\$ 177. After weeks of political maneuvering, the government finally managed to avoid a defeat, and the government’s proposal was finally passed on May 10th, backed by 306 out of 490 votes in the lower house, and 48 out of 69 in the Senate.

Four years later, in late April 2004, the Lula government announced a bill that raised the value of the minimum wage by R\$ 20, bringing it to R\$ 260, or just about US\$ 83 per month. The president’s proposal made it through the lower house, but on June 17th the Senate approved a larger rise, triggering a crisis for the government, which claimed it could not afford such an increase. The bill returned to the house on June 21st for a final vote, and on June 23rd the government rallied its legislators and by the sizeable margin of 272–172, rejected the Senate’s changes.

In 2000, higher W-Nominate ideal points were associated with greater probability of voting with the government for a smaller increase in the minimum wage, while in 2004 this relationship was reversed (Figure 1). Up to this point the results are compatible with an ideologically driven story: in 2000, those to the right of the scale (higher W-Nominate scores) voted with the center-right government, and in 2004, those to the left of the scale (lower scores) voted with the center-left government.

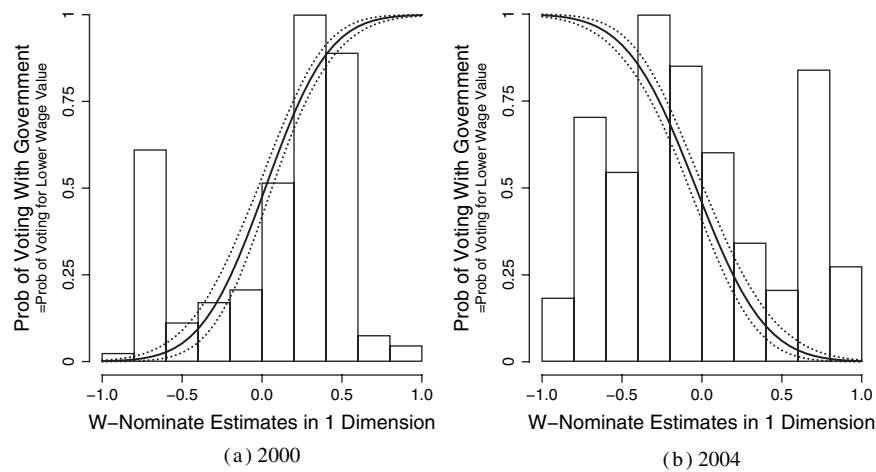
However, if one looks at the *content* of each side’s positions, it is clear that there is something besides ideology at work. While in 2000, the center-right government’s proposal called for a lower minimum wage than the opposition’s, as ideology would predict, in 2004 the center-left government’s proposal was *also for a lower* minimum wage than the opposition’s. In other words, after gaining power, the same PT that fought to raise the minimum wage beyond the government’s past proposals has fought attempts by the center-right opposition to raise the minimum wage beyond its own government’s proposals.

One can argue that when the matter reaches the final vote, legislators can be coerced or induced by the government and party leadership to vote a certain way, even if this is contrary to their own beliefs. During the earlier stages of the legislative process, however, there is a lot of space for cheap talk and position taking. While innocuous to the bill’s outcome, these earlier stages can help legislators save face with their constituents. For this reason, examination of the amendments presented to the minimum wage bills in each year can reveal more information about the legislators’s preferences.

A total of 55 and 79 amendments were presented to the government’s minimum wage proposals in 2000 and 2004, respectively. Not all of these amendments

⁶This dynamic is mostly true regarding measures that create or modify expenditure and revenue, which include most relevant policies. Nonetheless, there are some policies on which the positions have not changed, such as affirmative action and privatization. Further research will need to pin down exactly the extent of these “role contingent preferences” of political parties.

FIGURE 1 Final Votes on the Minimum Wage Bill (2000 and 2004)



Notes: Figures show the predicted probabilities of voting with the president on the final minimum wage vote (based on probit estimates) given the W-Nominate one-dimension ideal point estimate of legislators, the distribution of which is depicted in the histograms. 90% confidence intervals are also shown. The two roll calls used are identified as 2000108 and 2004055 in the Limongi & Figueiredo database.

proposed an actual value to the minimum wage; some were subscribed to by more than one legislator, some legislators made more than one proposal, and most were dismissed at early stages of the legislative process.⁷ With all these caveats in mind, the analysis of these amendments shows that in 2000, the correlation between estimated ideal points and the values of proposed amendments was a strong -0.56 , which means that legislators with lower ideal points (the putative “leftist” ones) proposed higher wages, behaving as ideologically motivated legislators should. In 2004, however the opposite holds. Legislators with higher estimated ideal points (“rightists”) proposed higher values for the minimum wage.⁸

The PT was not at ease with its new task. By 2004, its left-most faction had already split from the party

to form the PSOL, eight of the party’s remaining legislators subscribed to amendments that proposed increases, and at least nine voted against the government on May 10th. But, bluntly put, if one accepts that the position on the value of the minimum wage is an indicator of ideological preferences and believes that the W-Nominate estimates reveal the ideological left-right dimension, it would follow that the ordering for 2004 that is shown in Figure 1 is inverted. This is *not* to say that PT is a rightist party, but rather that ideology is not the main force driving the legislative votes.

Granted, legislators need not vote sincerely. The opposition, knowing it will be defeated, might simply take the more popular position, while the government, ultimately responsible for economic outcomes, can prefer a higher wage but know it is infeasible. These concerns might very well determine how legislators and government interact, but in any case they are *not* ideological. In fact, these votes provide some insight into what else (beyond ideology) influences how legislators behave, as there is considerable evidence that in both instances the government’s rallying of its base was accomplished through significant horse trading and resource distribution. In the 2000 vote, Pereira and Muller (2004) describe weeks of bargaining between the executive and its legislative base and note a considerable spike in the appropriations of funds to pay for legislator’s pet projects around the time of the vote. In 2004, the story is similar, with newspapers reporting a significant

⁷If a legislator presented or subscribed to more than one amendment, I took the average across all of the amendments he/she subscribed to. I dropped amendments that did not set a value to the minimum wage and the few amendments presented by Senators. When proposals called for staggered raises or more complicated formulas, I computed the average value of the proposal for the 12 month period starting on May 1 of the relevant year.

⁸Data on amendments was provided by the Senate Archives (SARQ/SEATEN). Correlations were computed between the value of amendments presented to the minimum wage bill and the one dimensional W-Nominate estimates of the ideal point of the proponent of the amendment. In 2000, there were 17 legislators proposing amendments, while in 2004 there were 46. Correlation coefficients were -0.56 and 0.13 , with p-values of 0.02 and 0.15 , respectively.

increase in appropriations around the time of the minimum wage vote with legislators that ultimately supported the government receiving more than 90% of those funds (Bragon 2004). In the rest of this paper I provide a framework to systematically separate between the role of ideology and the role of government provided incentives in shaping the behavior of legislators.

Ideology in the Brazilian Legislature

Few would claim that politicians in general, and legislators in specific, behave solely in ideological terms. In fact, many studies have shown that politicians pursue a varied set of goals, which include approving policies in line with their ideology, obtaining offices, and maximizing votes (Strøm and Müller 1999). Despite this relatively uncontroversial fact, it is frequently the case that legislative behavior is treated as if it were a direct indicator of ideology.⁹

This paper recognizes that observed legislative behavior is the end result of complex political processes. Ultimately, behavior is the product of preexisting ideological inclinations and of political exchanges made between the government and Congress. The theoretical framework employed here assumes that legislators have exogenously given propensities to support the president. These propensities are derived from ideological concerns and possibly also from electoral calculations. Even though legislators are clustered in parties, these propensities can vary individually, from the point of being indifferent to supporting the government in the absence of any political favors, to being extremely opposed to the president. The president, who initially holds all the resources, exchanges political favors for support by allocating resources either directly to legislators, or through political parties to obtain the necessary legislative support while spending the least resources.

Granted, the notion that the behavior comes *after* the political exchanges take place is slightly misleading, as the “giving-and-receiving” is, in reality, a continuous process. Nonetheless, the important point is that conceptually the W-Nominate estimated ideal points are the *result* of the president’s distribution of resources combined with underlying *ex-ante* preferences.

⁹An example of such confusion can be found in Alston and Mueller who present a model that is structured in terms of “policy positions” but then rely on W-Nominate scores as estimates of the “ideological position of each legislator and the president” (2006, 110).

After recognizing this fact, the obvious difficulty becomes obtaining some measure of ideology that is *exogenous* to legislative behavior. I accomplish this by relying on Timothy Power’s surveys of the Brazilian legislature for 1990, 1993, 1997, 2001, and 2005 (Power 2000). Of particular importance to the task at hand are the questions in these surveys that asked legislators to place *themselves and all other main parties* in the legislature on a left-to-right 10-point scale. It is important to note that placement questions such these capture perceptual differences among respondents, but also pick up undesired variation in response to the scale itself. Additionally, it is also possible that the meaning of the left-right scale that was used can vary over time. To deal with these problems, I estimate party positions indirectly from the data by accounting for legislator and year distortion effects.

Each survey yields a matrix of party placements and a vector of individual self-placements on a left-right ideological scale. Formally, let $P_{ij} \in [1, 10]$ be the placement in any of the surveys, of party j ($j = 1, \dots, M$) by legislator i ($i = 1, \dots, N$). The spatial model I use is simply:

$$P_{ij} = \alpha_i + \beta_i \pi_j + \varepsilon_{ij}, \quad (1)$$

where π_j is the “true” position of each party, α_i and β_i are legislator specific “shift” and “stretch” rescaling factors, and ε_{ij} is a well behaved disturbance term. Next, define $P_{ii} \in [1, 10]$ as the self-placement of legislators. Assuming that each legislator uses the same scale to place himself as he used to place all the other parties, each legislator’s placement onto the common scale (π_i) is defined as a simple linear transformation of the raw answer to the self-placement question (P_{ii}), as follows:

$$\pi_i = \frac{P_{ii} - \alpha_i}{\beta_i}. \quad (2)$$

It is straightforward to calculate π_i , once the parameters in equation (1) are estimated. However, as π_j has to be estimated along with α_i ’s and β_i ’s, this problem is akin to a regression *without* an independent variable, and consequently cannot be estimated directly by OLS. Instead, I approach the problem through a maximum-likelihood framework.¹⁰ Assuming

¹⁰Alternatively, one could adopt a principal components procedure (Aldrich and McKelvey 1977), which yields the exact same estimates (up to a linear transformation), but MLE should be more transparent to most readers and allows for the computation of the standard errors of the estimates.

the disturbance term is standard normally distributed, the probability of any observation is:

$$Pr(P_{ij}) = \phi\left(\frac{P_{ij} - \alpha_i - \beta_i \pi_j}{\sigma}\right) \frac{1}{\sigma}, \quad (3)$$

where ϕ is the standard normal density. The log-likelihood function to be maximized is then

$$\mathcal{L} = \sum_i \sum_j -\log(\sigma) - \frac{1}{2\sigma^2} (P_{ij} - \alpha_i - \beta_i \pi_j)^2. \quad (4)$$

The estimation procedure was conducted for each of the five surveys *individually*. A second rescaling procedure was used to make position estimates comparable across surveys. However, as the empirical tests shown later in the paper rely on the relative ideological distances between legislators and the president in any given year, it is not necessary to use estimates that are comparable over time.¹¹

It is worth mentioning here that the comparable estimates of ideology tell a story of a relatively stable ideological structure. The median legislator has shifted slightly to the left over time—probably due to the increase in the size of leftist parties—and there has been a slight decrease in polarization in recent years, caused mainly by the rightward drift of leftist parties. Overall, however, the left-right ordering of parties has been very persistent over time. These estimates make it possible to observe how ideology relates to legislative behavior and, more interestingly, whether one deviates from the other in any significant way. The punch-line here is that the changing patterns of legislative behavior described in the previous section cannot be accounted for by changes in ideology. Despite what their behavior indicates, ideology estimates suggest that the PT and the PDT are now closer than in the past, and there is no indication that the PL and PPB have moved to the left of the PSDB.

Figure 2 shows simple comparisons of the estimates of parties' ideology with ideal point estimates obtained via W-Nominate. Results are shown for each of the last five presidencies and are measured

relative to the president's position.¹² Collectively, the figures indicate a clear shift in the pattern of association between ideology and behavior in the legislature. During Collor's presidency (1990–92), there was an almost perfect association between behavior and ideology, but this association has weakened considerably over time. Ideology and behavior are far from unrelated, but there is clearly more than ideology going on.

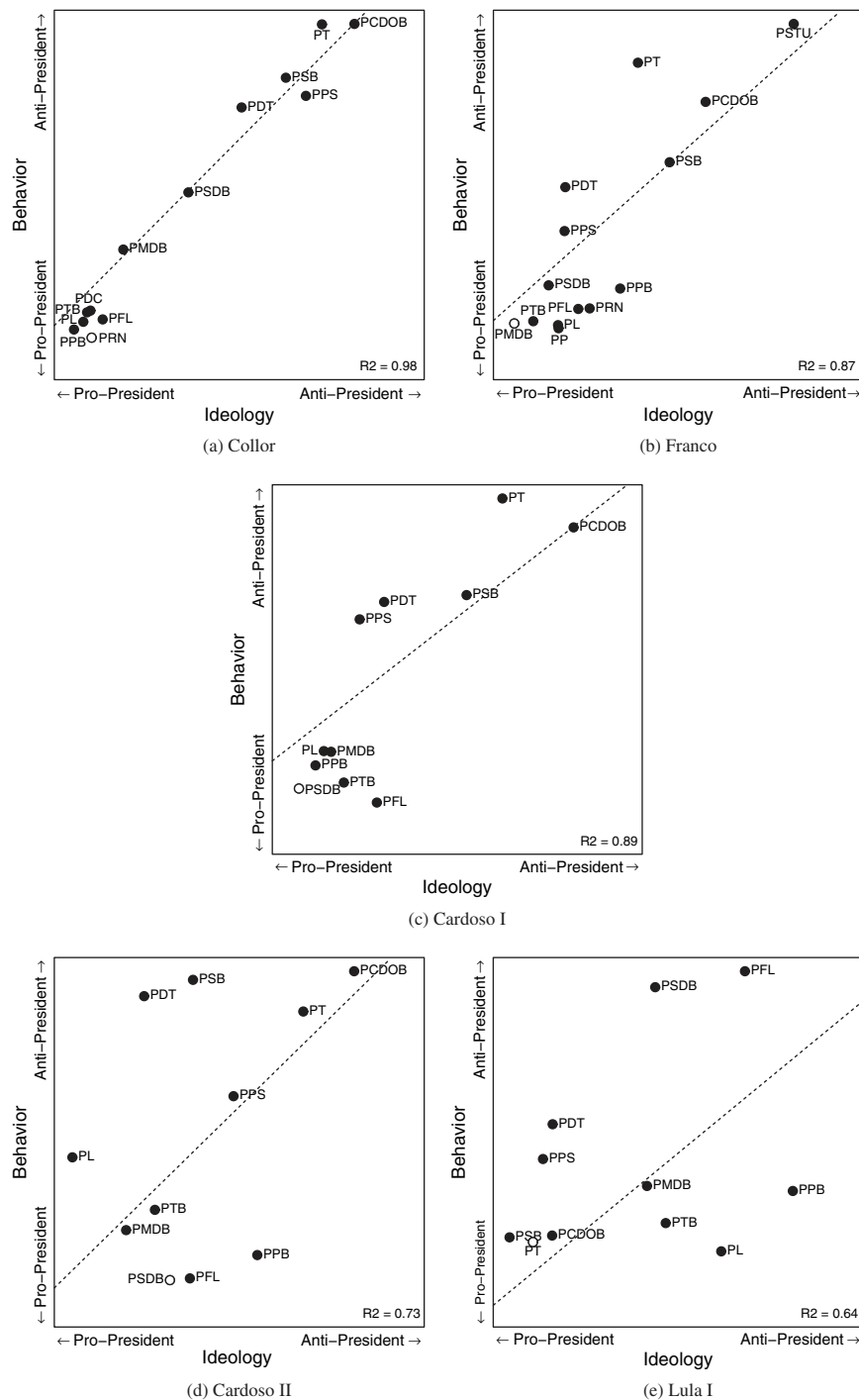
While this evidence suggests that the legislative behavior of parties cannot be accounted for by changes in ideology, the same cannot be said of the status of the parties vis-a-vis the president. The most general pattern in Figure 2 is that parties included in the cabinet tend to appear in the area of the graph below the regression line, reflecting the fact that when in the government parties are more supportive of the president than their ideology would suggest. Focus, for instance, on Figure 2(e), which refers to Lula's first term in office. The main adversaries of the government (PFL and PSDB) are clearly separated from the rest and are much further from the president in terms of behavior than ideology (especially the PSDB). The leftist parties that opposed the government during most of the period (PDT and PPS) are in a similar situation, albeit with less extreme positions. Lula's strange bedfellows (the rightist PL, PTB, and PP) are in the opposite situation and exhibit a behavioral pattern much more similar to the government's than predicted by ideology. The amorphous PMDB seems to keep its ideology, which probably reflects the fact that its *governista* wing cancels out the oppositionist wing. Finally, the behavior of the PT's "natural" allies (the leftist PSB and PC do B) is very close to the president both in terms of behavior and ideology.

This leads to the hypothesis that the underlying ideological alignment of parties is "scrambled" by the executive's distribution of resources, and for this

¹¹The details of this second rescaling procedure are available in the online appendix. For the complete set of estimates, and its analysis, see (Power and Zucco Jr. 2009).

¹²For behavior, I computed the average by party of the absolute distance between legislators W-Nominate ideal points and that of the government's leader in the *Câmara de Deputados*. For the ideology estimates, I computed the average by party of the absolute distance between respondent in the survey and the mean position of the president's party, but almost identical figures are obtained by using the party ideology estimates (π_j) instead. This poses a problem for Franco, who joined Collor's PRN prior to the 1989 elections, but left the party in May 1992, when the first serious corruption allegations against Collor emerged. He spent his whole presidency with no formal party affiliation and then joined the PMDB in 1997. Of the main parties that were part of his cabinet, the PMDB was the one *empirically* closest to the president and was the party used to produce Figure 2(b).

FIGURE 2 Ideology and Behavior Relative to the President: By Presidency



Notes: The horizontal axis is the average absolute distance between the party members' ideology estimates and the median legislator in the president's party. The vertical axis is the average absolute distance between the party members' W-Nominate ideal point estimates in one dimension and the ideal point estimate of the president's whip. Dotted line shows the regression without intercept of behavior on ideology.

reason the two measures do not always match. The ideologically structured picture that previous observers saw, I argue, was merely a coincidence that was caused by the overlap between government-opposition

cleavage, with the left-right ideological disposition of parties. During the Collor/Franco presidency, and during Cardoso's time in office, it was hard to distinguish between the effects of ideology and the

government's efforts to buy support. As is always the case, the president's actions induce a behavioral separation between the government and opposition camps, but with the presence of coherent coalitions, these efforts *reinforced* preexisting ideological differences. The result of this process was a snapshot of legislator behavior that suggested the existence of a much more ideologically polarized legislature than there really was.

Conversely, when the government's coalition is ideologically incoherent, a more selective scrambling of the underlying ideological disposition of parties occurs. In this scenario, the executive's actions *scramble* preexisting ideological differences by attenuating differences towards parties that are in the cabinet. Hence, the election of Lula—a nominally leftist president—and the formation of an ideologically incoherent coalition played the role of a natural experiment, allowing one to perceive *prima facie* evidence that behavior did not reflect ideology. If ideology were the “name of the game,” parties on the ideological right would not have changed their behavior and approached the PT, and parties on the left would not have moved away from the PT just because of their government or opposition status.¹³

Such outcomes are compatible with a theory of behavior in which legislators derive utility both from policy positions and from the fruition of political favors received from the president. While policy preferences are exogenously determined, the president can, and does, manipulate the provision of political favors to obtain the necessary legislative support. The president provides part of these favors to legislators directly in the form of pork, but another part is provided to the parties through posts in the cabinet. Now, such presence can contribute both to the provision of direct tangible benefits much alike pork, but also to less quantifiable but equally relevant political assets such as photo ops and credit claiming prerogatives. In this sense, being part of the government's ruling coalition is simply a proxy—though not the only one—for access to political favors.

¹³This begs the question of what causes coalitions to be more or less ideologically coherent. I analyze this issue in greater detail elsewhere, but here it suffices to say that presidents do not necessarily seek to form a coherent coalition, but rather the cheapest coalition. Moreover, it can be shown that it is frequently the case that the cheapest coalition is not necessarily a coherent one.

Explaining Legislative Behavior

Following up on this evidence, I now examine in greater detail the association between legislators' behavior, ideology, and the executive's use of resources to influence legislative voting. In the statistical analysis that follows, legislative behavior is measured from roll-call voting, and ideology is measured using the survey-based estimates described in the previous section. Following work done on executive-legislative relations in Latin America in general, and in Brazil in particular, I distinguish the provision of pork *to individual legislators* through the selective appropriation of funds (Ames 1987b; Pereira 2002, and others) from the allocation of control over parts of government *to parties* through the appointment of cabinet members (Amorim Neto 2006; Amorim Neto, Cox and McCubbins 2003; Geddes 1994, and others). Hence, the basic model around which the analysis is built can be conceptually summarized as:

$$\begin{aligned} \text{BEHAVIOR}_{it} = & \alpha_t + \beta_1 \text{IDEOLOGY}_{it}^* \\ & + \beta_2 \text{CABINET}_{jt} \\ & + \beta_3 \text{PORK}_{it} + \varepsilon_{it} \end{aligned} \quad (5)$$

where subscript *i* denotes legislator specific variables, *j* denotes the party specific variable, and *t* indicates that the data are observed yearly, from 1996 through 2006.¹⁴ A brief description of the variables used in the analysis follows, and more details are provided in the online supplement.

Data and Methods

BEHAVIOR is the absolute distance between the legislator's ideal point in one dimension and that of the president's “whip” (*Líder do Governo na Câmara*) in that same year. All ideal points were estimated using W-Nominate on yearly data,¹⁵ so estimates are constrained to lie in the $[-1, 1]$ interval, making absolute distances from the president lie in a $[0, 2]$ interval. The results discussed below also hold using an alternative operationalization of this variable, namely the frequency with which an individual legislator votes with the president, which is reported in the online appendix.

¹⁴I used budgetary data compiled by the *Câmara de Deputados*. Data for 2002 was unavailable.

¹⁵Only legislators that voted on at least 15 roll calls were included in the estimation.

Pork received by individual legislators (PORK) is operationalized as a legislator's success in getting his budget amendments implemented by the government in any given year.¹⁶ Brazilian legislators can propose a fixed number of individual amendments to the executive's budget proposal, which can add up to a previously determined ceiling, and include mostly infrastructure projects that benefit their constituencies. Since the budget law really only *authorizes* expenditures, the government is not obliged to actually spend the budgeted resources. Presidents can choose which amendments to carry out. Since legislators want their amendments to be implemented, selective use of the executive impounding power is an important tool to help secure support and discipline members of congress (Pereira & Muller 2004).

Party membership in the cabinet is operationalized as a dummy variable (CABINET), indicating whether the legislator's party was in the cabinet in a given year. Cabinet membership was observed at the beginning of each legislative session (February of each year). The relative few cases of parties changing cabinet affiliation during the year were ignored.¹⁷

The variable IDEOLOGY is the absolute distance of each legislator from the position of the mean legislator within the president's party. Since this variable is estimated from the Power surveys, it is measured once for each legislature rather than yearly (note the subscript t^* on this variable). Ideology estimates were not available for most individual legislators, so in the year-by-year regressions I used the mean ideological position for each party. While this is not an ideal solution, the main consequence of this strategy should be the addition of unbiased measurement error to the data, which should increase uncertainty in the estimates, but not necessarily introduce bias.

There is, however, one conceptual problem to this analysis. IDEOLOGY, like PORK, and in contrast to CABINET, should be an individual-specific variable. However, not all legislators answered the survey, and it was only possible to identify a few of those who did.

Of those, additional legislators were lost due to missingness in the PORK variable. The practical consequence is that there are only individual level ideology data for 14 to 34 legislators per year. For this reason, I pooled these identified yearly observations into a single data-set, which I refer to as the "reduced" pooled sample. Two important problems with these data jump immediately to the fore. First, it constitutes a severely unbalanced cross-sectional time-series,¹⁸ and for this reason it is impossible to explore the time structure in the data. Second, and perhaps more importantly given the nature of the data, it is at least possible that this subset of legislators is not representative of the whole sample. While there is a strong reason to suspect that the selection mechanism at work is quite random,¹⁹ I ensured against sample driven results by running the same model specifications in two other different samples.

One of these alternative samples consisted of a "full" pooled set with all legislators, simply using the party-mean ideology estimates as was done in the yearly regressions.²⁰ The other pooled sample was an imputed data set, where the mean party ideology values used in the full sample were replaced with imputed values for each legislator using the Amelia II procedure for missing data (King et al. 2001).²¹

One last issue is the possibility of endogeneity in the variable PORK. Note that this is not a problem with IDEOLOGY and CABINET. The former is conceptually fixed a priori, and while the latter can conceivably be influenced by legislative behavior, it is not something that is finetuned very often, and in practical terms is fixed for a relatively long period.

¹⁸The data consists of between one and 10 observations for 79 different legislators, for a total of 217 legislator/year observations.

¹⁹Even if the pool of legislators who answered the surveys were not random, the method for identifying legislators I used should be sufficiently randomizing, as it depends on a combination of over 15 variables that for the most part are not correlated with ideology in any obvious way.

²⁰This "full sample" includes 935 legislators, but as was the case with the reduced set, it is also severely unbalanced because most legislators move in and out of the sample very frequently. Many legislators only serve one term, or serve nonconsecutive terms. In addition, many legislators take leaves and are replaced by alternates, and others leave definitively to pursue other positions.

²¹More than 40 variables, including the average ideological position within each party, were used in this imputation. I imposed very high confidence priors so that the imputed values of ideology had to follow the distribution of ideology estimates for the legislature to which the year corresponded. I also constrained all legislator/year observations within a legislature to be the same, as was the case with the individual legislator estimates.

¹⁶Such data are only available for legislators that were serving in Congress in the preceding fiscal year. Hence, the N was generally lower in the first year of each legislature. It is worth noting that Lyne (2008) uses similar data as a measure of resources available to party leaders. Leaders do act as intermediaries in the resource distribution process, but authority over these resources ultimately lies with the government, and not with party leaders.

¹⁷In 1996, the PPB and the PPS joined in May. In 1999, the PPR joined the government in August. In 2005, the PPS left the cabinet in April, and the PP joined in August.

TABLE 2 Determinants of Legislative Behavior (Model 1)

	1996	1997	1998	1999	2000	2001	2003	2004	2005	2006
IDEOLOGY	1.011 0.038 <0.001	0.794 0.036 <0.001	0.531 0.034 <0.001	0.335 0.036 <0.001	0.327 0.035 <0.001	0.227 0.03 <0.001	-0.046 0.042 0.282	0.051 0.029 0.082	0.151 0.041 <0.001	0.117 0.022 <0.001
PORK	-0.199 0.035 <0.001	-0.246 0.043 <0.001	-0.219 0.042 <0.001	-0.093 0.04 0.022	-0.285 0.053 <0.001	-0.223 0.047 <0.001	0.012 0.069 0.867	-0.044 0.038 0.239	-0.243 0.05 <0.001	-0.006 0.031 0.836
CABINET	0.594 0.088 <0.001	0.609 0.082 <0.001	-0.004 0.076 0.963	-0.044 0.077 0.566	-0.118 0.075 0.116	-0.165 0.063 0.008	-0.493 0.114 <0.001	-0.101 0.065 0.124	-0.537 0.093 <0.001	-0.124 0.048 0.01
IDEOLOGY $\times$ CAB	-0.906 0.089 <0.001	-0.893 0.082 <0.001	-0.152 0.077 0.049	-0.289 0.054 <0.001	-0.389 0.057 <0.001	-0.161 0.047 0.001	0.16 0.064 0.013	-0.117 0.033 <0.001	-0.135 0.046 0.004	-0.214 0.027 <0.001
Const.	-0.408 0.051 <0.001	-0.132 0.055 0.017	0.015 0.05 0.764	0.424 0.066 <0.001	0.629 0.067 <0.001	0.48 0.059 <0.001	0.638 0.098 <0.001	0.495 0.07 <0.001	1.021 0.092 <0.001	0.558 0.044 <0.001
N	387	393	474	286	452	388	230	377	332	364
ADJ R ²	0.78	0.74	0.61	0.62	0.69	0.62	0.16	0.36	0.74	0.61

Notes: Dependent variable is BEHAVIOR, a measure of legislative voting in which lower values mean greater proximity to the president's positions. Regressions estimated by OLS. Standard errors and p-values are shown below estimates.

PORK, however, is handed out throughout the year, even though considerable amounts are disbursed in December—the end of the fiscal year. Pereira and Muller (2004) show that pork disbursements can be both a reward and an enticement for legislators, and for this reason they affect and are affected by the legislator's voting record. To deal with this issue, I estimated my pooled-data models by two-stage least squares, using individual measures of seniority and experience as instruments for pork. Details of the estimation and its results are provided in the online appendix, but results are essentially the same as those described below, with the exception that PORK is not statistically significant in all models.

Results

The first set of results are shown in Table 2, where Model 1 is fit to year-by-year data-sets. Results for the pooled data analysis are reported in Table 3. Model 2 has the same specification as Model 1, except for the inclusion of year effects. Models 3 and 4 explore the time trend, the former with the inclusion of a linear measure of time and an interaction of that measure and ideology, and the latter with the inclusion of an interaction between ideology and year dummies. Results for these last three models are reported for the reduced, full and imputed samples.

In broad terms, the three components of the model (ideology, handouts to parties, and handouts to legislators) explain a considerable portion of the variation in the dependant variable. Each play relevant roles in most years, with the expected effects: greater ideological distance is associated with more behavioral distance from the president; and greater success in obtaining pork, and the party's presence in the cabinet are associated with more proximity to the president in terms of behavior.

The general consistency of the results is, nonetheless, subject to one remark: the results for 2003 are very atypical in the year-by-year regressions and exhibit a considerable drop in the explanatory power of the model. I return to this issue in the next section, as I discuss alternative interpretations of the general results. For now, however, I draw attention to two other very striking features that emerge from the data analysis, namely that the substantive impact of ideology on behavior *decreases* markedly over time, and that ideology has considerably *smaller* effects for parties *in* the cabinet.

With respect to the latter, one can immediately note the negative coefficients in the interaction between ideology and presence in the cabinet (Models 1 and 2). This interaction term also implies that presence in the cabinet has a greater effect on the behavior of legislators that are more distant ideologically from the president. The net effect of ideology for parties in the government coalition is typical either zero or very

TABLE 3 Determinants of Legislative Behavior: Pooled Data Analysis

	Reduced Sample			Full Sample			Imputed Sample		
	Mod 2	Mod 3	Mod 4	Mod 2	Mod 3	Mod 4	Mod 2	Mod 3	Mod 4
IDEOLOGY	0.432 0.074 <0.001	0.278 0.047 <0.001	0.367 0.076 <0.001	0.874 0.037 <0.001	0.552 0.021 <0.001	0.789 0.037 <0.001	0.832 0.035 <0.001	0.567 0.021 <0.001	0.757 0.036 <0.001
IDEOLOGY × CAB	−0.173 0.038 <0.001			−0.229 0.014 <0.001			−0.237 0.014 <0.001		
IDEOLOGY × TIME		−0.032 0.006 <0.001	‡		−0.057 0.002 <0.001	‡		−0.056 0.002 <0.001	‡
PORK	−0.263 0.067 <0.001	−0.225 0.068 0.001	−0.296 0.07 <0.001	−0.168 0.015 <0.001	−0.157 0.015 <0.001	−0.176 0.015 <0.001	−0.2 0.014 <0.001	−0.179 0.015 <0.001	−0.208 0.015 <0.001
CABINET	−0.171 0.066 0.01	−0.395 0.042 <0.001	−0.412 0.041 <0.001	−0.088 0.021 <0.001	−0.37 0.01 <0.001	−0.384 0.01 <0.001	−0.017 0.02 0.398	−0.309 0.01 <0.001	−0.317 0.009 <0.001
TIME	†	0.047 0.009 <0.001	†	†	0.07 0.003 <0.001	†	†	0.065 0.003 <0.001	†
Const.	0.274 0.105 0.01	0.458 0.086 <0.001	0.405 0.105 <0.001	−0.241 0.041 <0.001	0.105 0.027 <0.001	−0.097 0.041 0.019	−0.237 0.039 <0.001	0.068 0.026 0.009	−0.108 0.04 0.007
N	0.59	0.49	0.55	0.61	0.5	0.58	0.54	0.44	0.51
ADJ R ²	231	231	231	3675	3675	3675	4225	4225	4225

Notes: † Year effects were included. ‡ Interactions of IDEOLOGY with dummies for years were included. Estimates were obtained through OLS, and standard errors and p-values are shown below estimates. In all models, year effects and interaction terms were jointly significant with *p*-value <0.001. Dependent variable is BEHAVIOR, in which lower values indicate greater proximity to the president's position. In the "full sample," IDEOLOGY is averaged by party.

small, even though the effect of ideology on behavior is almost always clearly positive for those not in the cabinet. Parties lose their ideological considerations when they are part of the president's cabinet.

As for the decline of ideology over time, the trend can be spotted in the sheer size of the ideology coefficients in the year-by-year regressions and also in the interaction between the linear measure of time (TIME) and ideology in Model 3. Estimates for the interaction term between ideology and the year effects (Models 3) confirm the trend and are reported in Figure 3(b).

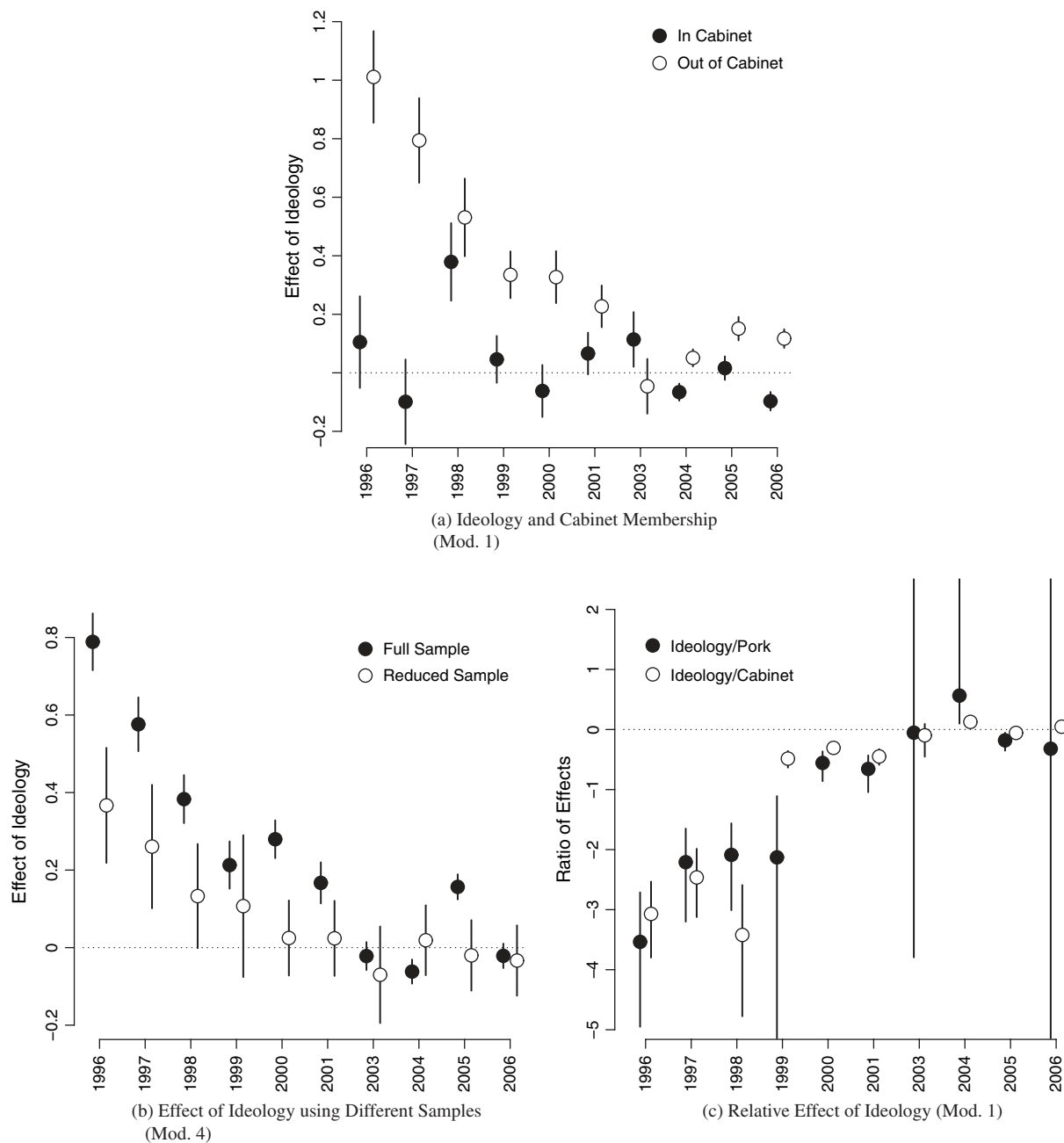
As both of these results involve the interpretation of interaction terms, they are much better observed graphically. The left panel shows not only the secular declining trend in the effect of ideology, but also that ideology matters more for the behavior of parties that are out of the cabinet. The center panel depicts the cumulative effect of ideology as estimated by Model 4,

both in the reduced sample and with the full sample. The figures based on the imputed sample and on the two-stage least squares estimates on the full sample are essentially the same, albeit with larger confidence intervals.

In particular, Figure 3(b) shows that in substantive terms, this is a very sharp decline indeed. In 1996, a one standard deviation increase in ideological distance from the president caused an increase of between 0.9 to 1.3 standard deviations on the behavioral distance scale. Since then, this effect has basically vanished. The inference to be made is that the role of ideology has declined for Congress as a whole and that it has never mattered much for legislators whose parties are in the cabinet. An interesting corollary is that resources that are distributed to parties do influence how individual legislators behave.

Overall, the results are supportive of the idea laid out earlier that the executive's distribution of

FIGURE 3 Effects of Ideology on Legislative Behavior: 1996–2005



Notes: Figures (a) and (b) show the point estimates and the 95% confidence intervals for the effects of ideology over time, as estimated by Models 1 and 4 respectively. Results for the latter case are shown for the “Full” and the “Reduced” samples. Figure (c) shows the ratio between the coefficients on IDEOLOGY and those on PORK and CABINET, respectively, as estimated by Model 1. Bootstrapped 90% confidence intervals about these estimates are shown.

resources creates a selective scrambling of the underlying ideological organization of the legislature. The results also indicate that ideology, resources distributed to parties, and those distributed to legislators matter in determining legislator behavior, but that their relative weights vary from year to year, probably

reflecting changes in strategy and the executive’s political capacity. Results suggest that ideology has lost importance in recent years, which is a robust finding that can also be spotted if its effects are measured *relative* to the other components of the model, as shown in Figure 3(c). The *ratio* between the

size of the effects of ideology relative to party and individual handouts have tended to zero over time, suggesting that ideology has become less important not only in absolute terms, but also *vis-a-vis* other determinants of legislative behavior.

Method to the Madness?

Ideology is not the sole determinant of legislative behavior, and moreover, its importance has declined over time. However, under Lula *all* three basic components of my framework—ideology, party goods, and individual perks—exhibit smaller effects than in previous periods. Consequently, based on the very same results, one could reasonably claim that the whole structure of legislative-executive relations—and not just the role of ideology—changed between Cardoso and Lula. The idea here would be that there is nothing really structural in how the executive and legislative branches relate to each other, but rather that each president implements his own *modus operandi* according to his style, strengths, and the relevant issues of his time.

In 2003, and to a lesser extent in 2004, the clear patterns of the Cardoso period are blurred, and the explanatory power of the model is considerably lower than in other years. This suggests that at least in the beginning Lula's term, cabinet positions and pork played a smaller role in determining behavior. The question is whether this was a temporary disruption that can be accounted for by a new president who is learning on the job or whether a new president implies a new game. The last two years of data tend to support the former alternative, but they are in no way definitive, as PORK is not statistically significant in 2006. At this point, it is still impossible to say with certainty whether executive-legislative relations in Brazil have shifted to a different pattern, or whether Lula's second term will reveal a return to the previous "normality." There are other pieces of evidence, however, that suggest that the hypothesis of a complete lack of structure is implausible.

For instance, some trends in the data began to show before Lula became president, and therefore cannot be attributed to his personal style. In fact, evidence shown in Figure 2 suggests an increasing disjunction between ideology and legislative behavior beginning with Collor, and the downward trend in the importance of ideology is clear in the regressions results *throughout* the Cardoso period.

Qualitative evidence contributes to the picture very emphatically. It would be hard to deny that

Lula's first term was disastrous in terms of coalition management. By many accounts, Lula began his government by handing too many resources to the PT (Franco, Vasconcelos, and Lima 2005) and failed to accommodate the interests of key allies. The government handled Congress very poorly on a regular basis, and failed in critical moments, as was seen in a series of mishaps in the run up to the election of backbencher Severino Cavalcanti to the key position of Speaker of the House. Things seem to have changed in Lula's second term, which started in 2007. Since his reelection, he personally took up the task of dealing with parties, has given more attention to "allies," and put more effort into building and managing a broader and more solid coalition. Thus, plenty of signs hint at a return to a relationship between Congress and the presidency similar to the one that existed under Cardoso.

Finally, the lower explanatory power of PORK and CABINET early in Lula's term can be attributed in part to the probable existence of the infamous *mensalão*, an alleged scheme to exchange outright bribes for legislators' support. If the government was, in fact, handing cash to certain legislators in return for their votes, this alternative to "regular" exchanges with Congress could reduce the importance of traditional currencies such as cabinet positions.

Until time reveals how Lula and Congress will behave in the next few years, or until the data series can be extended further back in time, it is impossible to decide between the two hypotheses. Nonetheless, all things considered, the hypothesis of a temporary disruption of rather stable patterns of executive-legislative relations combined with a secular declining trend in the role of ideology appears more plausible than the absence of a stable pattern in executive-legislative relations.

Though it seems clear that ideology has become less important, my argument is that it was never the sole determinant of legislative behavior. The main reasons why previous observers concluded that parties behaved ideologically was not only because ideology was more important in the past, but rather because the government's resource distribution strategies can reinforce or attenuate the underlying ideological cleavage. In the earlier period, being in or out of government overlapped with being on the right or on the left, so the government's efforts furthered exacerbated ideological differences. In the latter period, the government picked coalition partners from all over the political spectrum, producing the scrambled behavioral images shown in this paper. While in both periods being in the coalition and

receiving pork brings legislators and parties closer to the president, a snapshot of their end results is considerably different.

Conclusion

Observed legislative behavior is not always a good way to measure legislators' ideology, especially when political exchanges between the executive and the legislative, as shown here, influence the behavior of legislators. This paper develops an exogenous measure of ideology and shows that legislative voting departs from it in systematic ways. In the process, it documents the importance of the executive's spoils distribution strategy in multiparty presidential systems. Observers have long believed that Latin American presidents influence legislative voting, but the techniques I have developed enable us to assess how important this influence really is. For Brazil, it reveals an intriguing decline in the role of ideology as a determinant of legislative behavior during the last 10 years.

The paper contributes directly to the debate about the functioning of the Brazilian political system. It shows that the relative stability of the ideological alignment of postdemocratization Brazilian parties contrasts with the changes in what some parties seem to stand for in the legislature. The innovative combination of pork, cabinet positions, and an exogenous measure of ideology in a single framework makes it possible to show that legislators' behavior is influenced *both* by what they receive individually and by what their party receives from the government. These findings represent a synthesis between the depictions of the Brazilian Congress as the arena of the locally minded, pork seeking, free-floating legislator (Ames 1987a), and the competing view that internal rules result in a legislature structured around parties, which behave in roughly ideological terms (Figueiredo & Limongi 2002). Nonetheless, this should be qualified as "weighted synthesis," for two reasons. First, while parties do seem to matter, they do not seem to matter in an ideological way. When parties are included in the cabinet, their members' vote with the president regardless of ideology. This finding suggests that parties matter because they help mediate the distribution of resources by the president, even though not all exchanges of support for presidential handouts are made through parties.

Furthermore, the decline in the role of ideology identified in this paper suggests that whatever party

organization exists in Congress, it is not ideological one, which contradicts at least part of the account by Figueiredo and Limongi (2002). Previous observations of ideological behavior by parties, I argue, were driven by the fact that there was a coincidence between the left-right and the government-opposition dimensions that projected the appearance of an ideologically organized legislature. The shift in the makeup of the government coalitions that occurred with the election of Lula helped make clear that ideology cannot fully explain legislative behavior. This paper dwells on this evidence and suggests that access to government resources mattered as much then as it does now. Though our capacity to distinguish between the government and the ideological effects is limited during the earlier period, ideology was never the sole determinant of legislative voting.

The paper also shows that presidents can greatly influence how legislators behave. Though it relies on data from Brazil, we would expect to see similar patterns of presidential influence whenever presidents exercise unilateral control over important political resources while lacking a disciplined majority in the legislature. Wherever this is the case, the conditions are set for the exchange of pork and patronage for votes to matter at least as much as ideology. By distributing resources, presidents cause actual voting patterns in Congress to diverge from what would be predicted solely by ideological predispositions. In these cases, understanding executive-legislative behavior requires understanding the processes by which political favors are exchanged. This implies that spatial models of legislative voting based solely on ideology present an incomplete picture of how policy is made.

Although evidence from other countries is less systematic, it suggests that ideological inclinations do not match actual legislative behavior in a number of other Latin American countries. Presidents, through the use of state resources under their control, exert an almost irresistible attraction that helps structure legislative behavior along pro- and antigovernment lines. For example, the almost 20 years of executive dominated politics in Bolivia—known as the *rodillo oficialista*—exhibit these same characteristics. Nominally minority presidents were able to maintain ideologically heterogeneous coalitions to support economic liberalization mainly through the distribution of pork and patronage, leading to votes in Congress that were frequently split along government-opposition lines, even though the composition of each side varied considerably (Mayorga 2006). In Uruguay, the link

between legislative behavior and pro- or antigovernment status has been documented (Morgenstern 2001), and so has the fact that two factions of the same party will vote against each other if one is in and the other out of government (Buquet, Chasquetti, and Moraes 1998). In Ecuador, too, voting has frequently been determined by the commitment to support the government, and not because of the merits of any specific piece of legislation. Such commitments are often established through short-lived and quickly shifting deals, sometimes public, sometimes secretly established between the president and parties, or even informal groups of legislators (Mejía Acosta 2004). Notwithstanding the current lack the data to extend the analysis in this paper to other countries, the observations of other researches suggest that the processes analyzed here also occur in other presidential systems.

The most general point of this paper is that under certain circumstances one should not attempt to infer ideology directly from observed legislative voting patterns without a careful look at the processes that lead up to these votes. This point transcends the case of Brazil, and that of presidential systems in general and is potentially applicable to the analysis of legislative behavior in general. As Spirling and McLean (2006) have shown, particular aspects of executive-legislative relations under the Westminster system can lead to legislative voting patterns that are not correctly accounted for by simple models of roll-call analysis. The lesson to be learned is that if one is to really make sense of observed legislative behavior, it is necessary to take into consideration the political processes that generate the patterns that are ultimately observed. In this sense, the use of an exogenous measure to separate ideology from other incentives faced by legislators, such as the one presented here, could be applied in any setting in which survey data on legislators' attitudes and roll-call votes exist.

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